

Assignment 3: Motion Planning and Control

Autonomous Driving Systems - Planning and Control Module

Salvatore Di Lorenzo

Matricola: 207689

AA 2025/2026

Contents

1	Exercise 1: Static obstacle avoidance at low speed	2
2	Exercise 2: Static obstacle avoidance at high speed	4
3	Bonus exercises	8
3.1	Fernet Planner's execution time optimization	8
3.2	Highest speed achieved	9

Introduction

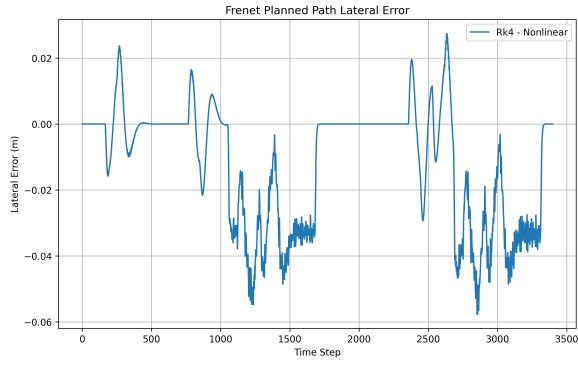
This assignment focuses on the implementation and validation of the local motion planning algorithm "Optimal Trajectory Generation in a Frenet Frame" for autonomous vehicles operating in dynamic environments. The primary objective was to integrate the planning module in the already developed pipeline, meaning in planning and control algorithm to work together, on the same mathematic representation of the vehicle through the non-linear Single Track model.

1 Exercise 1: Static obstacle avoidance at low speed

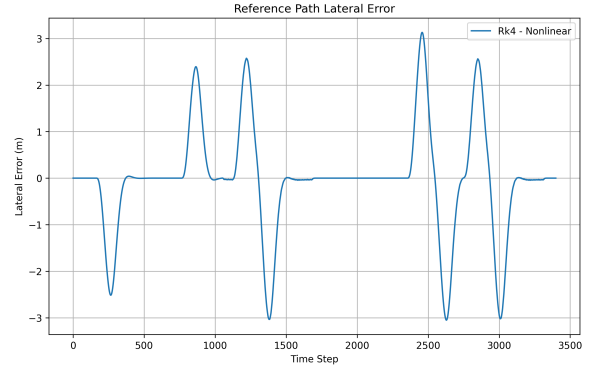
This section presents the results of the low-speed simulation, conducted at a speed of 10 m/s and then 15 m/s. For the control layer, the Pure Pursuit algorithm was used in the simulations. Please note that, in both the first and the second exercise, the Frenet planner's parameters were left unchanged; more precisely, they had the following values (parameters that were not used in this assignment have been omitted):

$$\begin{aligned} Roadwidth_{max} &= 5.0, & Roadwidth_{delta} &= 0.5, \\ DT &= 0.2, & MAX_T &= 5.0 & MIN_T &= 4.5, \\ K_{jerk} &= 0.1, & K_{time} &= 0.1, & K_{deviation} &= 1.0, \\ K_{lateral} &= 1.0, & K_{longitudinal} &= 1.0 \end{aligned}$$

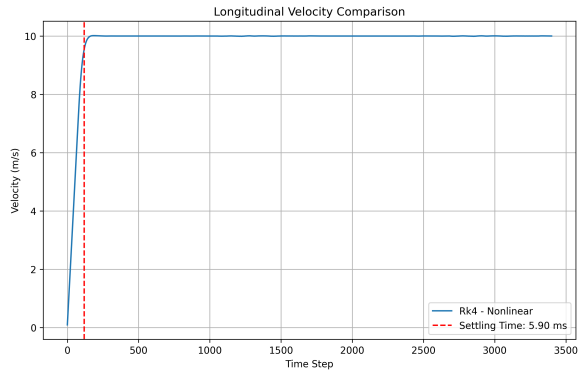
At both speeds, the planner built a trajectory that the control layer was able to follow very accurately (Fig. 1, 2)



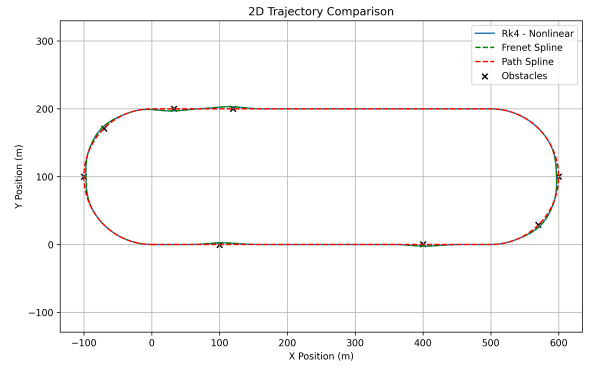
(a) Lateral Error w.r.t Planned Path



(b) Lateral Error w.r.t Reference Path

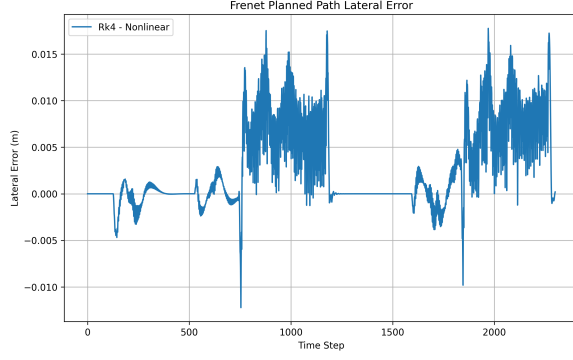


(c) Longitudinal Velocity

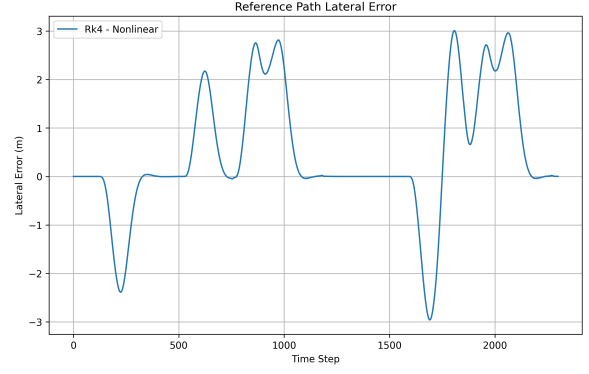


(d) Trajectory

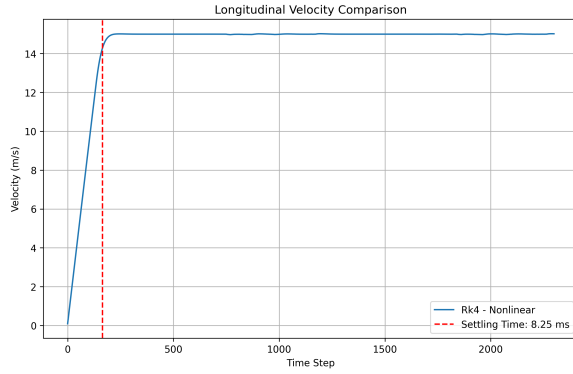
Figure 1: Frenet Planner and Pure Pursuit Controller at 10 m/s



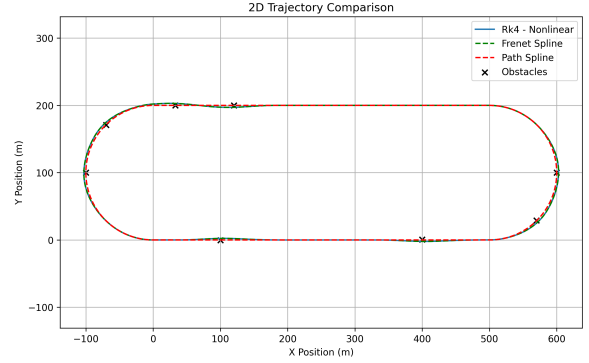
(a) Lateral Error w.r.t Planned Path



(b) Lateral Error w.r.t Reference Path



(c) Longitudinal Velocity

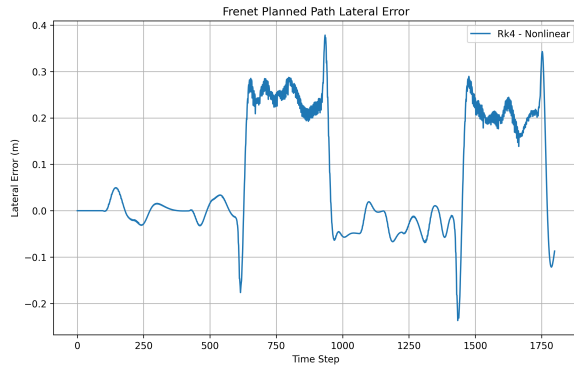


(d) Trajectory

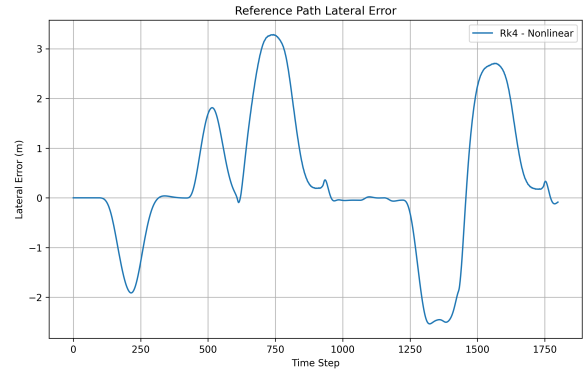
Figure 2: Frenet Planner and Pure Pursuit Controller at 15 m/s

2 Exercise 2: Static obstacle avoidance at high speed

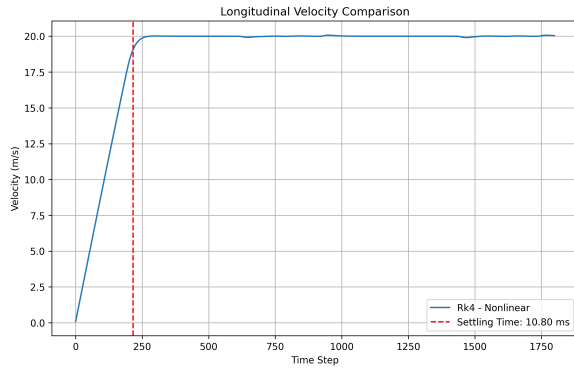
This exercise focused on testing the planner (and controllers) at higher speeds (i.e., 20 m/s and 25 m/s). Firstly, at 20 m/s, Pure Pursuit and MPC with a kinematic internal model were used. In this case, both controllers managed to successfully follow the planned path. However, due to the limited nature of the kinematic MPC, which does not account for lateral forces, the tracked path produced by Pure Pursuit presented significantly less lateral error with respect to the planned path (Fig. 3, 4).



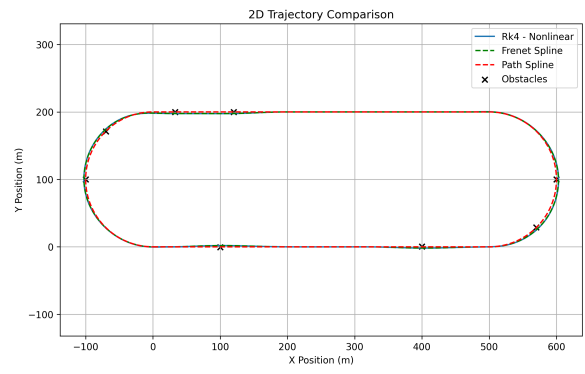
(a) Lateral Error w.r.t Planned Path



(b) Lateral Error w.r.t Reference Path



(c) Longitudinal Velocity



(d) Trajectory

Figure 3: Frenet Planner and Kinematic MPC Controller at 20 m/s

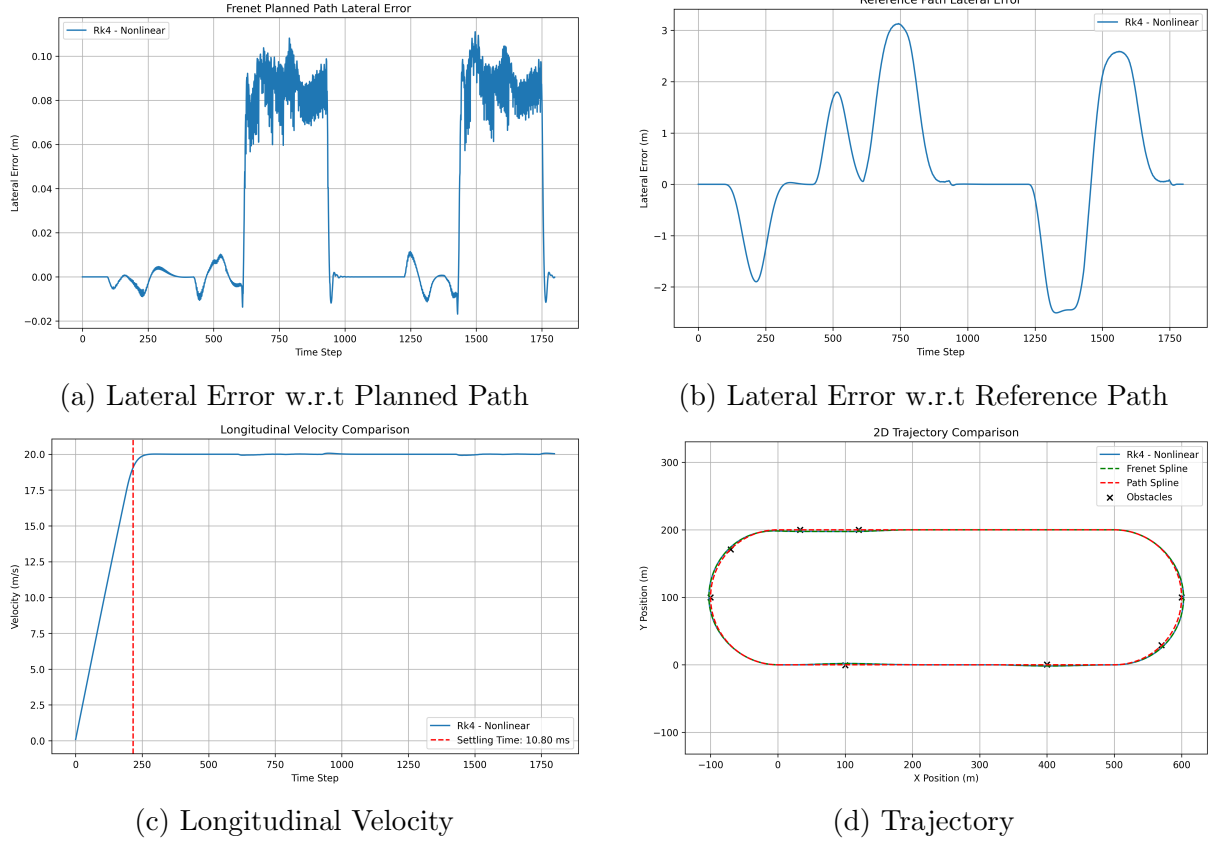


Figure 4: Frenet Planner and Pure Pursuit Controller at 20 m/s

For the 25 m/s tests, every developed controller was used. More precisely, the kinematic MPC managed to complete an entire lap at this speed, although it was at its limit, with the lateral error reaching 1 m (Fig. 5).

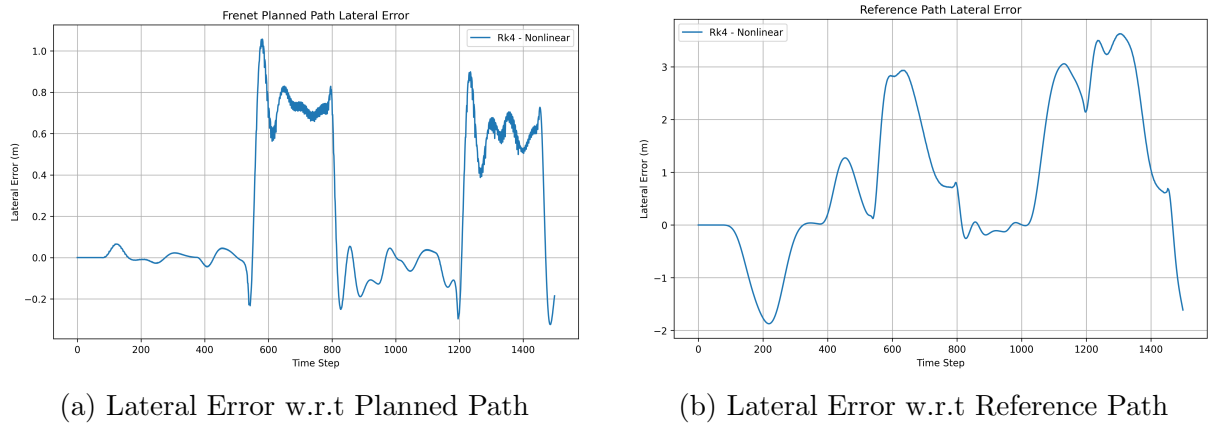
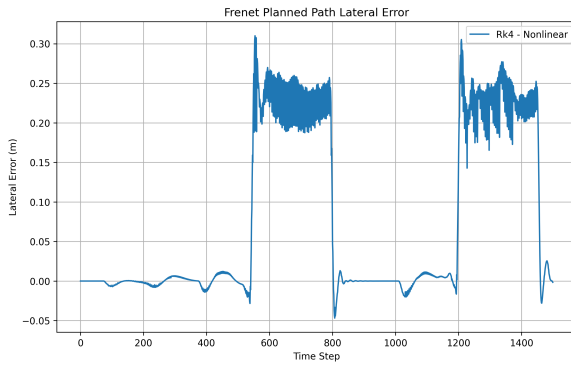


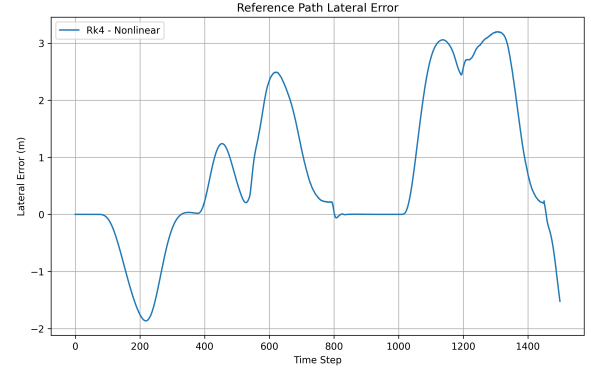
Figure 5: Frenet Planner and Kinematic MPC Controller at 25 m/s

The other three controllers performed consistently with the analysis from the previous assignment. Pure Pursuit exhibited stable behavior but slow convergence to the planned path, resulting in a greater lateral error than Stanley, whose drawback was instead frequent oscillations (Fig. 6, 7). MPC with a Linear Single-Track model proved to be the

best in terms of both stability and mean lateral error, despite its peak error being the highest (Fig. 8).

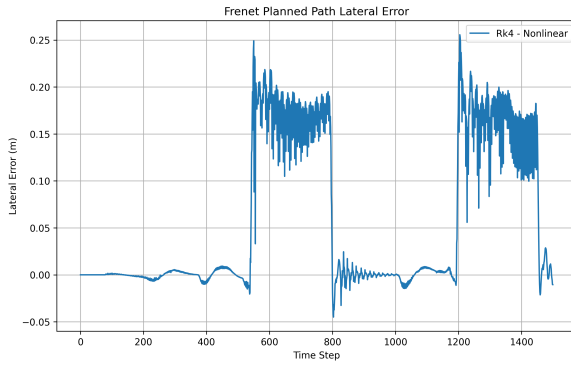


(a) Lateral Error w.r.t Planned Path

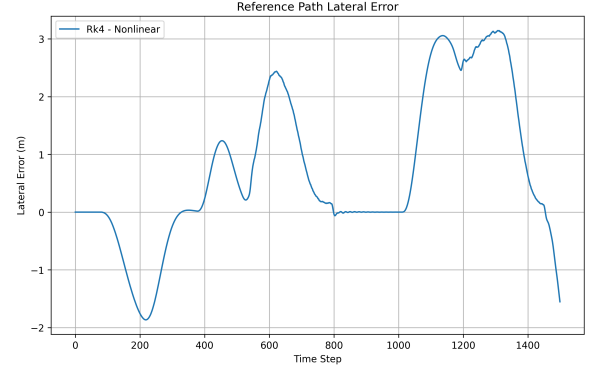


(b) Lateral Error w.r.t Reference Path

Figure 6: Frenet Planner and Pure Pursuit Controller at 25 m/s

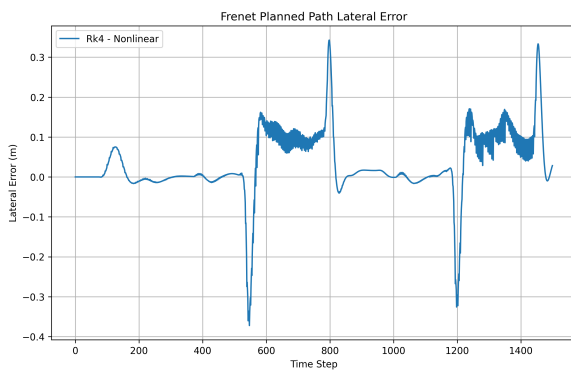


(a) Lateral Error w.r.t Planned Path

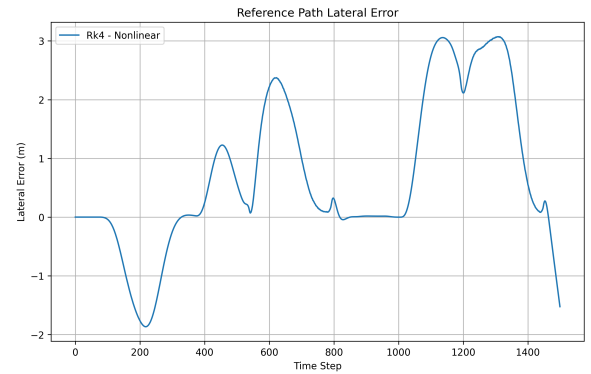


(b) Lateral Error w.r.t Reference Path

Figure 7: Frenet Planner and Stanley Controller at 25 m/s



(a) Lateral Error w.r.t Planned Path



(b) Lateral Error w.r.t Reference Path

Figure 8: Frenet Planner and Linear MPC Controller at 25 m/s

Please note that each of the controllers successfully tracked the planned path, respecting the maximum lateral error constraint on the reference path and maintaining a

stable velocity after the settling time (5% max variation, as described in the previous assignment). The trajectory and velocity plots have been omitted from this report for brevity.

3 Bonus exercises

In this exercise the Frenet parameters were slightly modified in order to achieve two different optimization: execution time and maximum achievable speed.

3.1 Frenet Planner's execution time optimization

The objective of this section is to explain the reasoning behind the changes made to the Frenet planner in order to reduce its execution time. First of all, the unchanged version of the planner registered a mean execution time of about 30 ms (Fig. 9).

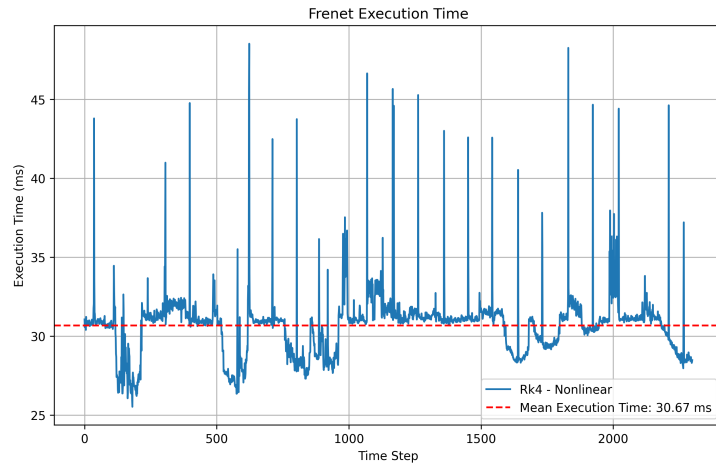


Figure 9: Frenet Execution Time - no optimization

To achieve a reduced execution time, the goal was to minimize the number of paths the Frenet planner had to choose from. For this reason, the **road width** parameter was reduced to 4.0 m (since the constraint does not allow for lateral errors greater than 4.0 m). Furthermore, since these tests are conducted in a constant speed scenario, the path length can be adapted as well: it is not strictly necessary to have different lengths for the generated paths; a single length is sufficient. After some tests, both **MAX_T** and **MIN_T** values were reduced to generate a single beam of paths, all having the same length. Please keep in mind that the selected value cannot be too high, as the path would result in an inaccurate prediction, nor can it be too small, because some time is still needed for the controller to take an alternative route to avoid an obstacle without forcing the vehicle into unfeasible maneuvers. The results of the tuning are the following parameters:

$$\begin{aligned}
 Roadwidth_{max} &= 4.0, & Roadwidth_{delta} &= 0.5, \\
 DT &= 0.2, & MAX_T &= 4.5 & MIN_T &= 4.45, \\
 K_{jerk} &= 0.1, & K_{time} &= 0.1, & K_{deviation} &= 1.0, \\
 K_{lateral} &= 1.0, & K_{longitudinal} &= 1.0
 \end{aligned}$$

This enhanced Frenet planner was tested along with the Pure Pursuit controller at 15 m/s, resulting in a tracking performance very similar to the non-optimized version (Fig. 2b 10a). While the planned path was almost identical, the mean execution time was nearly one-fourth of the original one (Fig. 10b). The same planner was tested in combination with the MPC linear controller as well, which successfully respected the constraints up to a maximum speed of 30.3 m/s, without any substantial difference from the non-optimized version (Fig. 11, 13b).

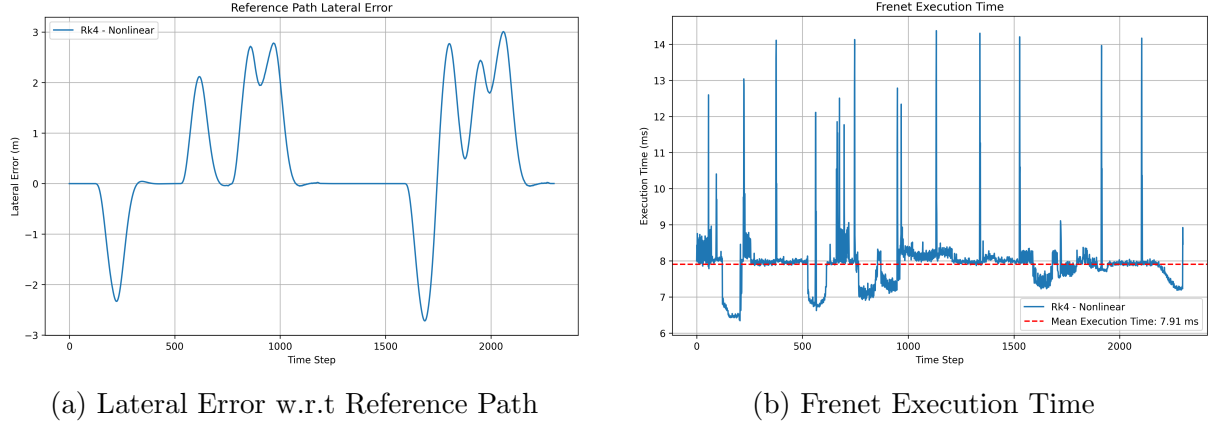


Figure 10: Optimized Frenet Planner and Pure Pursuit Controller at 15 m/s

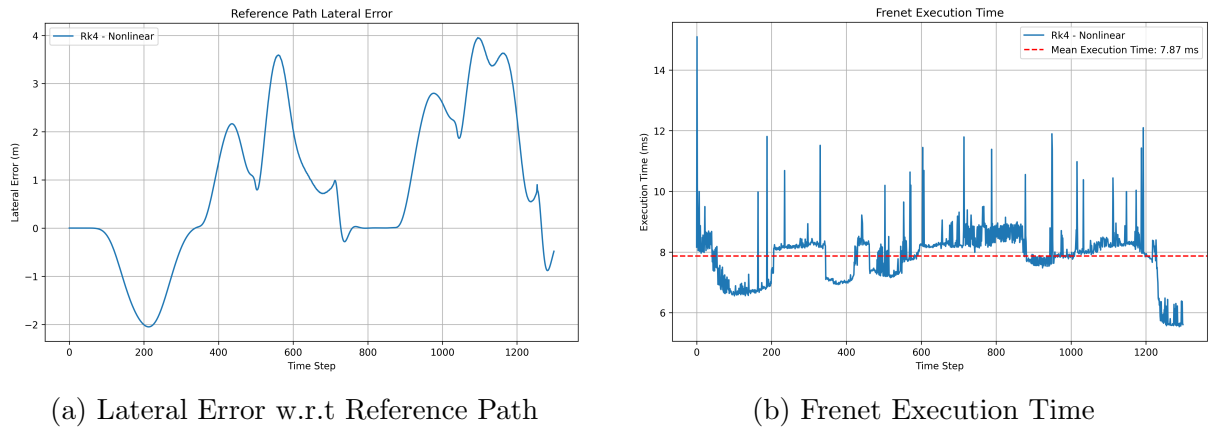


Figure 11: Optimized Frenet Planner and Linear MPC Controller at 30.3 m/s

3.2 Highest speed achieved

In this part of the assignment, the objective was to push the controller to its limit, reaching the highest possible speed while respecting the constraints.

For this purpose, each controller was tested, except for the kinematic MPC due to its inherent limitations. The achieved results were almost identical to the maximum speeds reached in the previous assignment; for this reason, plots are omitted except for the linear MPC, which achieved the highest speed in this case as well (Fig. 12). Please note that this speed was achieved leaving the Frenet parameters unchanged.

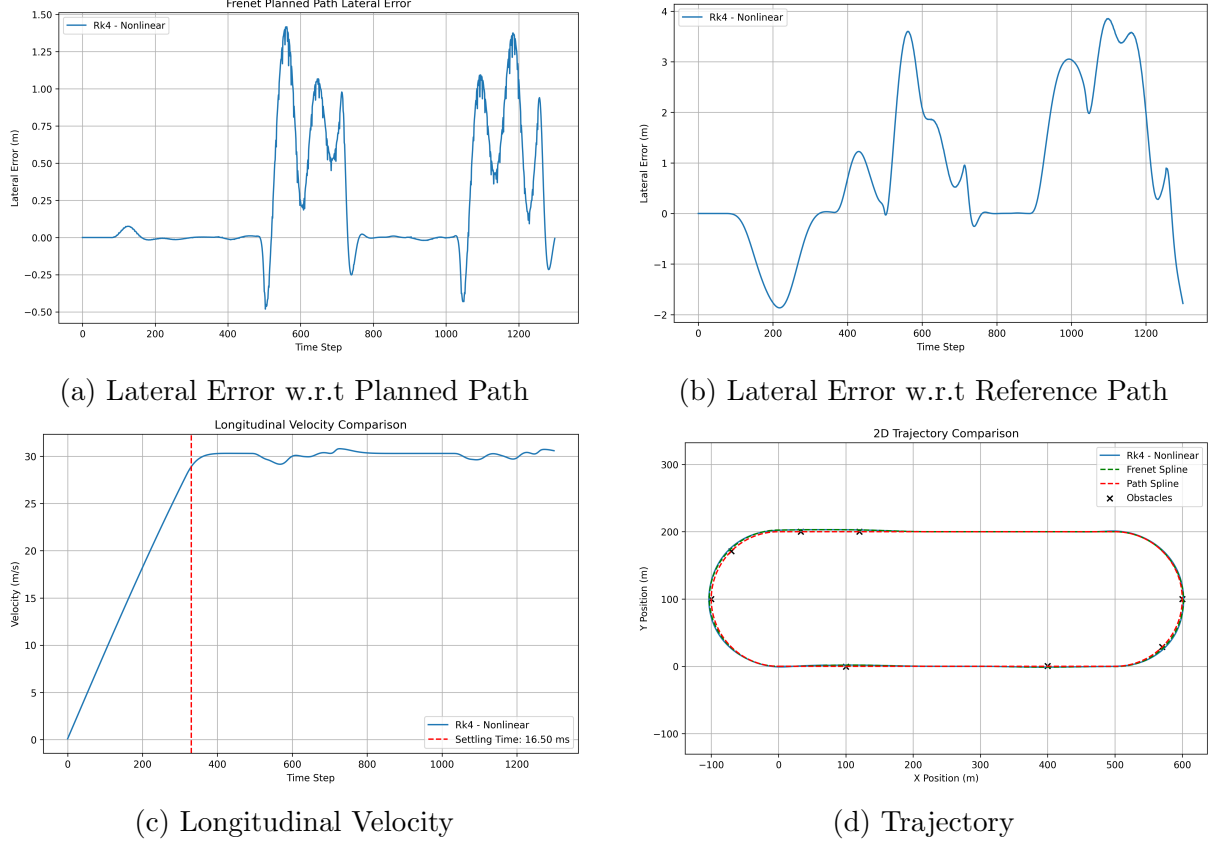
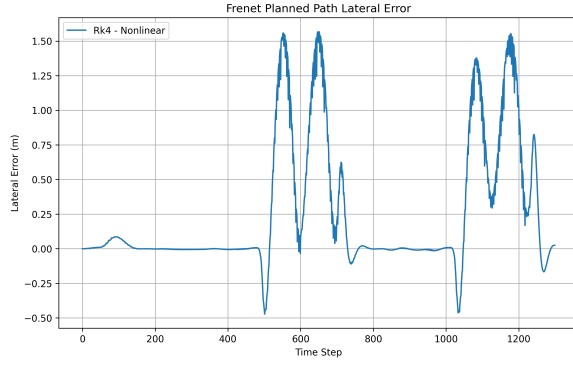


Figure 12: Frenet Planner and Linear MPC Controller at 30.3 m/s

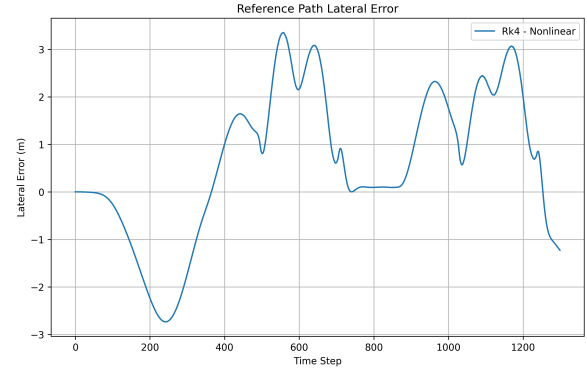
Since this limit was controller-bound, to achieve a greater speed, the objective became smoothing the path generated by Frenet. This was done by modifying some of its parameters. First of all, a wide beam of paths is not needed since, at such high speeds, a deviation of several meters from the reference path would be unfeasible in a short time; this is why the **road width** was reduced to 3.5 m. On the other hand, the **road width delta** was decreased to allow for a finer-grained choice among the paths. The cost function was updated as well: the **jerk cost** was increased to promote smoother paths, and the **deviation cost** was decreased to give the vehicle slightly more freedom during maneuvers. The results of these changes are the following parameters:

$$\begin{aligned}
 Roadwidth_{max} &= 3.5, & Roadwidth_{delta} &= 0.2, \\
 DT &= 0.2, & MAX_T &= 4.5 & MIN_T &= 4.45, \\
 K_{jerk} &= 0.5, & K_{time} &= 0.1, & K_{deviation} &= 0.5, \\
 K_{lateral} &= 1.0, & K_{longitudinal} &= 1.0
 \end{aligned}$$

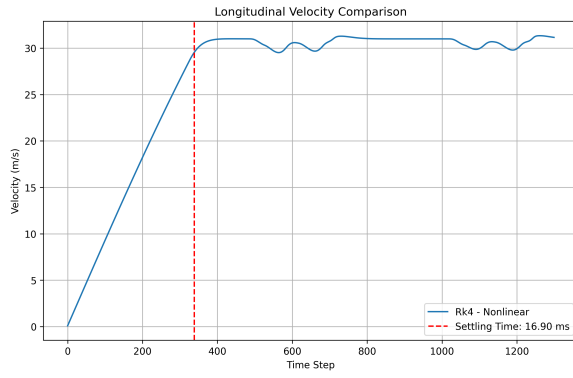
This combination allows for smoother curves, and with these changes, the vehicle reached a maximum speed of 31 m/s while respecting both constraints (Fig 13).



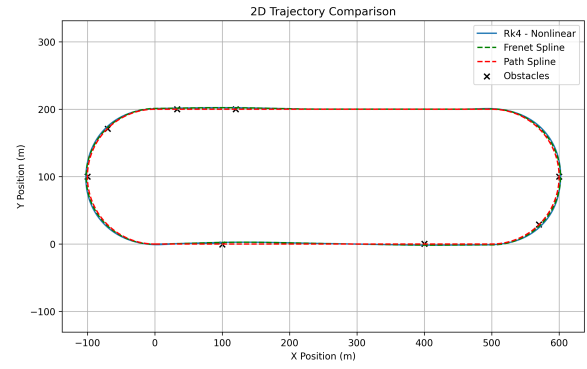
(a) Lateral Error w.r.t Planned Path



(b) Lateral Error w.r.t Reference Path



(c) Longitudinal Velocity



(d) Trajectory

Figure 13: Optimized Frenet Planner and Linear MPC Controller at 31 m/s